ENGR-095
Claire Volinski
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During this week in ENGR-095, we began our first real exploration into circuitry- specifically the connection between hardware and software. This week, we started with building our websites. Admittedly, I had a lot of difficulty getting acquainted with Github for the first time. I wasn't familiar with its layout- and quite frankly confusing and- (in my opinion) overdesigned UI. For being an easy coding program it was a bit of a learning curve.

However, despite this, I was able to get a handle on the site, and by the end of the next lab on Wednesday I was quite happy with how my site turned out.

However, beyond just setting up the site, we learned more about the workflow of creating software, and integrating it with the Arduinos we were given on Wednesday, after more practical and long learning on Monday. We learned about the flow of software- the blocks of actions, and the diamonds that indicated decisions- along with the fact there are other symbols- "with" statements, as well as various other symbols seen in the slideshow diagrams that we ourselves did not quite touch on within the module itself. In the next lab session, we learned about the pieces of code- variables, the sleep timers, and managed to follow the code guidelines and make our own blinking light Arduino. We were able to code specifically for the exact pin output- using the values True/False, as well as the 0 and 1 for the "on" and "off" aspects of the LED itself- which we applied via the code. Granted, the process was simple, as we were generally following the instructions, but I found it incredibly fascinating.

In fact, throughout only the first few weeks of this course, I have found a very distinct love for electronics and circuitry, of which I was not aware I enjoyed as much as I did before. Having my background in Electricity and magnetism from my high school has really helped me conceptualize circuits in a new way. I have found a real interest in robotics and more hands-on construction and machining- which is a little out of the realm of my bioengineering major- which presents some interesting considerations regarding my major.

However, besides just the Arduino and the flowchart introduction, I really enjoyed the hands-on-nature of this week- it was a lot more satisfying than working with online models of circuits I would use frequently in my high school. I have found myself really enjoying the concepts and practice of simply creating code to physical responses, as well as the engineering aspects of how these are incorporated into machines- although we have not done that yet, mixing the manufacturing and coding is something I am very much interested in and looking forward to!

So far, I have had an incredible time not only this week, but in the module itself! It is incredibly helpful and inspiring what I've learned so far.